Business Summary Report: Predictive Insights for Collections Strategy

1. Summary of Predictive Insights

Based on the EDA (Task 1) and the predictive modeling plan (Task 2), the following insights were observed:

- Customers under 30 years old with two or more missed payments and credit utilization above 50% show significantly higher risk of delinquency.
- High debt-to-income ratio (> 0.5) consistently correlates with late or missed payments.
- Customers with short account tenure (< 12 months) tend to have weaker repayment behavior.

Key Insights Summary Table:

Key Insight	Customer Segment	Influencing Variables	Potential Impact
Customers with high credit utilization and frequent missed payments are at highest risk of delinquency.	Under 30 years old, 2+ missed payments, high credit utilization (> 50%)	Missed Payments, Credit Utilization, Age, Debt-to-Income Ratio	Targeted outreach could reduce 30-day delinquency rates by over 10% in this segment.

2. Recommendation Framework

Restated Insight:
Customers under 30 with 2+ missed payments and credit utilization above 50% are at significantly high risk of delinquency.

Proposed Recommendation:
Launch a 6-week SMS outreach campaign targeting these customers to reduce 30-day delinquency by at least 10%.

Specific:
Target customers under 30 with 2+ missed payments and high credit utilization.

Measurable:
Track reduction in 30-day delinquency; goal is a 10% decrease.

Actionable:
Use existing SMS infrastructure for outreach.

Relevant:
Aligns with Geldium's goal to lower delinquency and improve repayment behavior.

Time-bound:
Campaign will run for 6 weeks.

Justification and Business Rationale:
This targeted campaign is simple, cost-effective, and data-backed. It directly reaches a high-risk segment with an intervention that is easy to scale. By focusing on this group, Geldium can achieve significant impact in reducing overall delinquency while maintaining a positive customer experience.

3. Ethical and Responsible AI Considerations

- Potential Bias: The model may under-represent or misclassify certain demographic groups if historical data is skewed (e.g., income or location-based bias). Fairness checks are necessary.
- Explainability: Random Forest provides strong predictive performance by identifying complex patterns in customer financial behavior. Feature importance analysis also helps explain which variables contribute most to delinquency risk.
- Responsible Decision-Making: Model outputs are reviewed by humans before action. The SMS campaign is supportive, not punitive.
- Other Ethical Principles: Transparency is maintained through explainable features. Privacy is ensured by using anonymized data. Ongoing monitoring ensures the model adapts to changing customer behavior.